User, Group, Authentication Hardening Report

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Preparation

Initial Setup and Baseline Audit

Setup Steps

1. Pull and Run the Docker Environment

The first step was preparing the Ubuntu 22.04 assessment environment using Docker.

Pull the Container
docker pull nwendlo1/c1w2m6

Run the Container
docker run -it --rm --name lynis-ir-lab nwendlo1/c1w2m6

This created an isolated environment for performing the security audit and remediation tasks.

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2. Update Package Lists

Before installing additional tools, package repositories were updated.

apt update

This ensured the latest package metadata and security updates were available.

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3. Install Lynis

Lynis was installed using the Ubuntu package manager.

apt install lynis -y

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4. Verify Lynis Installation

The installation was verified by checking the version.

lynis --version

This confirmed that Lynis was installed successfully and ready for auditing.

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5. Conduct the Initial Baseline Audit

A full system audit was performed to establish the baseline security posture.

lynis audit system

This generated:

- Security warnings
- Hardening suggestions
- Vulnerability information
- Hardening index score

Baseline Security State

Initial Hardening Index

Example Initial Score

Hardening index : 58 [############]

The initial score showed that the Ubuntu system required significant hardening improvements.

Key Areas of Concern

1. Weak Authentication Policies

Lynis identified weak password aging settings and authentication controls.

Risk

Weak authentication increases the likelihood of:

- Brute-force attacks
- Credential compromise
- Unauthorized access

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2. Insecure SSH Configuration

SSH allowed insecure settings such as root login.

Risk

Attackers could gain direct administrative access remotely.

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3. Missing Firewall Protection

The system firewall was disabled or improperly configured.

Risk

Unrestricted network access exposed the system to external threats.

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4. Improper File Permissions

Sensitive files had insecure access permissions.

Risk

Unauthorized users could potentially access authentication information.

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5. Outdated Packages

Several installed packages required updates.

Risk

Known vulnerabilities could be exploited by attackers.

Policy Alignment

Key Security Policies Relevant to the Audit

1. Least Privilege Policy

Users and services should only have the minimum permissions required.

Importance

This policy guided actions related to:

- File permissions
- User accounts
- Group management

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2. Secure Authentication Policy

Authentication systems must enforce strong password and login controls.

Importance

This policy influenced:

- Password aging configurations
- SSH hardening
- Root login restrictions

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3. Patch Management Policy

Systems must remain updated with current security patches.

Importance

This policy guided package update and vulnerability remediation actions.

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4. Network Security Policy

Systems must restrict unnecessary network exposure.

Importance

This policy directed firewall configuration and SSH restrictions.

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5. System Hardening Policy

Servers must follow CIS Benchmark recommendations for secure configurations.

Importance

This policy provided the framework for implementing Lynis recommendations and improving the hardening score.

Detection and Analysis

Identification of Vulnerabilities

Vulnerability 1: Weak Password Policies

Evidence

Lynis reported insufficient password aging configurations.

Verification Command
grep PASS_MAX_DAYS /etc/login.defs

Findings

Password expiration settings were weak or missing.

Policy Violation

This violated the company’s secure authentication policy.

Vulnerability 2: Insecure SSH Configuration

Evidence

Lynis flagged insecure SSH settings.

Verification Command
grep PermitRootLogin /etc/ssh/sshd_config

Findings

Root login was enabled.

Policy Violation

This violated the least privilege and remote access security policies.

Vulnerability 3: Disabled Firewall

Evidence

Firewall protection was missing or inactive.

Verification Command
ufw status

Findings

The firewall was inactive.

Policy Violation

This violated network security requirements.

Vulnerability 4: Improper File Permissions

Evidence

Sensitive authentication files had weak permissions.

Verification Command
ls -l /etc/shadow

Findings

Permissions were more permissive than recommended.

Policy Violation

This violated the least privilege policy.

Vulnerability 5: Outdated Packages

Evidence

Lynis identified outdated packages missing security updates.

Verification Command
apt list --upgradable

Findings

Several packages required updates.

Policy Violation

This violated patch management policies.

Prioritization of Vulnerabilities

Highest Priority Vulnerabilities

1. Improper File Permissions

This vulnerability was prioritized first because exposure of authentication files could lead to complete system compromise.

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2. Insecure SSH Configuration

Root SSH access presented a high risk of remote administrative compromise.

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3. Disabled Firewall

Without network filtering, the system remained exposed to unauthorized connections.

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4. Weak Password Policies

Weak authentication controls increased the risk of credential attacks.

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5. Outdated Packages

Unpatched vulnerabilities could eventually be exploited if left unresolved.

Containment, Eradication, and Recovery

Remediation Actions

Remediation 1: Strengthening Password Policies

Strategy

Improve authentication security by enforcing password aging controls.

Actions Taken

nano /etc/login.defs

Updated Settings

PASS_MAX_DAYS 90
PASS_MIN_DAYS 7
PASS_WARN_AGE 14

Alignment with Policies and CIS Benchmarks

- Supports secure authentication policy
- Aligns with CIS password aging recommendations

Remediation 2: Hardening SSH

Strategy

Restrict remote administrative access.

Actions Taken

nano /etc/ssh/sshd_config

Updated Settings

PermitRootLogin no

Restart SSH

systemctl restart ssh

Alignment

- Enforces least privilege
- Aligns with CIS SSH hardening standards

Remediation 3: Enabling Firewall Protection

Strategy

Restrict unnecessary inbound traffic.

Actions Taken

ufw allow OpenSSH
ufw enable

Alignment

- Supports network security policy
- Aligns with CIS firewall recommendations

Remediation 4: Correcting File Permissions

Strategy

Protect sensitive authentication files.

Actions Taken

chmod 640 /etc/shadow
chmod 644 /etc/passwd

Alignment

- Enforces least privilege
- Aligns with CIS file permission recommendations

Remediation 5: Updating Packages

Strategy

Install security updates and patches.

Actions Taken

apt update && apt upgrade -y

Alignment

- Supports patch management policy

- Reduces exposure to known vulnerabilities

Post-Incident Activity

Verification and Comparison

Post-Remediation Hardening Index Score

Initial Score

Hardening index : 58

Final Score

Hardening index : 78

Improvement

The system showed a significant increase in overall security posture after remediation.

Comparison and Analysis

Improvements Observed

Authentication Security

Password aging policies improved account security and reduced long-term credential exposure.

SSH Security

Disabling root login reduced the likelihood of unauthorized remote administrative access.

Network Security

Firewall activation restricted unnecessary access to system services.

File Security

Permission hardening improved protection of sensitive authentication files.

System Maintenance

Package updates reduced exposure to publicly known vulnerabilities.

Remaining Issues and Recommendations

Remaining Issues

1. Multi-Factor Authentication Not Configured

MFA was not implemented during the lab.

Recommendation

Deploy MFA for administrative accounts to strengthen authentication security.

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2. Intrusion Detection Not Installed

No intrusion detection solution was configured.

Recommendation

Install tools such as:

- AIDE

- Fail2Ban
- OSSEC

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3. Centralized Logging Missing

Logs remained stored locally only.

Recommendation

Implement centralized logging and SIEM integration.

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4. Automated Auditing Not Configured

Security auditing remained manual.

Recommendation

Schedule automated Lynis audits using cron jobs.

Clear and Organized Reflection on the IR Process

Reflection on Each Phase

Preparation

The preparation phase established a secure and organized audit environment. Conducting the baseline audit helped identify the system's current weaknesses before remediation began.

This phase emphasized the importance of:

- Preparation
- Documentation
- Security baselines

Without a baseline, it would have been difficult to measure improvements later.

Detection and Analysis

This phase improved my understanding of:

- Linux vulnerabilities
- Authentication weaknesses
- Access control issues
- Network security risks

One challenge was determining which Lynis findings required immediate attention versus lower-priority recommendations.

I addressed this by prioritizing vulnerabilities based on:

- Potential impact
- Exposure level
- Company policy alignment

Containment, Eradication, and Recovery

This phase demonstrated how practical hardening actions reduce risk significantly.

The remediation strategies were effective because they:

- Restricted access
- Reduced attack surfaces
- Improved authentication security
- Strengthened system integrity

One challenge involved editing configuration files carefully without breaking services.

I overcame this by:

- Verifying changes before restarting services
- Following CIS Benchmark guidance

Post-Incident Activity

The post-remediation audit validated the effectiveness of the security improvements.

This phase highlighted the importance of:

- Verification
- Continuous monitoring
- Ongoing maintenance

Comparing hardening scores clearly demonstrated measurable improvement.

Overall Reflection

This lab reinforced how the Incident Response process provides a structured methodology for identifying, prioritizing, and remediating security weaknesses.

Key lessons learned include:

- Security auditing is essential for proactive defense.
- Hardening significantly improves system resilience.
- Verification is critical after remediation.
- Policies and benchmarks provide valuable guidance for secure configurations.

The structured IR approach can be applied in real-world environments for:

- Compliance audits
- Server hardening
- Vulnerability management
- Security incident response
- Cloud security operations

This lab also improved my practical experience with:

- Lynis
- Linux administration
- Authentication hardening
- Firewall management
- Security best practices

These skills are highly valuable for DevOps, cybersecurity, and cloud security roles where maintaining secure infrastructure is critical.